

Chapter 3 - Rings and Moduls of Fractions

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Exercise 1

(i) ($\Rightarrow$): Let M be generated by $\{m_1, m_2, \dots, m_n\}$, for $\forall i \in [1, n] \cap \mathbb{Z}$, we have

$$\forall r, t \in S, \exists s \in S, \text{s.t. } m_i s(r - t) = 0, \text{i.e. } m_i s = 0$$

Denote the s above as s_i . Since M is finitely-generated, we can let

$$s = s_1 s_2 \dots s_n$$

And have $sM = 0$.

(ii) ($\Leftarrow$): When such s exist, then for $\forall \frac{m_1}{s_1}, \frac{m_2}{s_2} \in S^{-1}M$,

$$s(s_1 m_2 - s_2 m_1) = 0 \implies \frac{m_1}{s_1} = \frac{m_2}{s_2}$$

So $S^{-1}M = 0$.

□

Exercise 2

By *Proposition 1.9*, the problem is equivalent to prove that for any $\frac{a}{s} \in S^{-1}A$, we have

$$\forall \frac{b}{t} \in S^{-1}A, 1 - \frac{a}{s} \frac{b}{t} \text{ is a unit in } S^{-1}A$$

Since $1 + \mathfrak{a}$ is a multiplicative set, $\exists c \in \mathfrak{a}, \text{s.t. } st = 1 + c$. Now

$$\begin{aligned} 1 - \frac{a}{s} \frac{b}{t} &= 1 - \frac{ab}{st} \\ &= 1 - \frac{ab}{1+c} \\ &= \frac{1+c-ab}{1+c} \end{aligned}$$

And we can easily see $\frac{1+c}{1+c-ab} = 1 + \frac{ab}{1+c-ab} \in S^{-1}A$. So the inverse of $1 - \frac{a}{s} \frac{b}{t}$ is $1 + \frac{ab}{1+c-ab}$.

□

Exercise 3

Define

$$f : (ST)^{-1}A \longrightarrow U^{-1}(S^{-1}A)$$

by mapping

$$\frac{a}{st} \longmapsto \frac{\frac{a}{s}}{\frac{t}{1}}$$

We'll prove f is an isomorphism:

(i) Surjection: Trivial.

(ii) Injection:

$$\frac{\frac{a_1}{s_1}}{\frac{t_1}{1}} = \frac{\frac{a_2}{s_2}}{\frac{t_2}{1}}$$

$\Leftrightarrow$

$$\exists t \in T, \text{ s.t. } \frac{t}{1} \left(\frac{a_1 t_2}{s_1} - \frac{a_2 t_1}{s_2} \right) = 0$$

$\Leftrightarrow$

$$\exists t \in T, \text{ s.t. } t(a_1 t_2 s_2 - a_2 t_1 s_1) = 0$$

$\Leftrightarrow$

$$\frac{a_1}{s_1 t_1} = \frac{a_2}{s_2 t_2}$$

So f is injective.

(iii) Homomorphism: Trivial.

□

Exercise 4

Define

$$f : S^{-1}B \longrightarrow T^{-1}B$$

by

$$\frac{b}{s} \longmapsto \frac{b}{f(s)}$$

We'll prove it's an isomorphism:

(i) Homomorphism: Trivial

(ii) Surjection: Trivial

(iii) Injection:

$$\frac{b_1}{f(s_1)} = \frac{b_2}{f(s_2)}$$

$\Leftrightarrow$

$$\exists s \in S, \text{ s.t. } f(s) (b_1 f(s_2) - b_2 f(s_1)) = 0$$

$\Leftrightarrow$

$$\frac{b_1}{s_1} = \frac{b_2}{s_2}$$

□

Exercise 5

(i) By *Proposition 3.11-(v)*, the operation S^{-1} commutes with the formations of radicals. So $\forall \mathfrak{p} \subseteq A$,

$$(\text{Nil}(A))_{\mathfrak{p}} = \text{Nil}(A_{\mathfrak{p}}) = 0$$

Therefore by *Proposition 3.8*, $\text{Nil}(A_{\mathfrak{p}}) = 0$.

(ii) No. Counterexample given here

Exercise 6

(i) Σ is nonempty, therefore has maximal elements by *Zorn's Lemma*.

- (ii) (I) When $S \in \Sigma$ is maximal, we'll prove $A \setminus S$ is a minimal prime ideal. $\forall x, y \in A \setminus S_0$, let

$$S_x = \{sx^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}, S_y = \{sy^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$$

Since $S \in \Sigma$ is maximal and $S_x, S_y \supsetneq S$, we have $S_x, S_y \notin \Sigma, 0 \in S_x, 0 \in S_y$. Therefore $\exists s_1, s_2 \in S, n_1, n_2 \in \mathbb{Z}^+$, such that

$$s_1 x^{n_1} = s_2 x^{n_2} = 0$$

And we can see

$$s_1 s_2 (x + y)^{n_1 + n_2 - 1} = 0, s_1 s_2 (xy)^{n_1 n_2} = 0$$

Which is to say $x + y, xy \in A \setminus S$, and we can easily obtain $A \setminus S$ is an ideal. The prime property is trivial, and the maximal property can be obtained directly from (II) and the maximal property of S .

- (II) When $A \setminus S$ is a minimal prime ideal of A : Denote $I = A \setminus S$. For $\forall x, y \in S$, we have $x, y \notin I$. If $xy \in I$, by the prime property of I we will immediately obtain $x \in I$ or $y \in I$, which is a contradictory. So $xy \notin I, xy \in S$, S is a multiplicatively closed subset. Also, if S is not maximal in Σ , the one in Σ strictly contains S will correspond to a smaller prime ideal in $A \setminus S$, which contradicts with the minimal property of I .

□

Exercise 7

- (i) (I) ($\Leftarrow$): When $A \setminus S$ is a union of prime ideals, $x \in A \setminus S$ or $y \in A \setminus S$ will immediately implies $xy \notin S$. So $xy \in S$ will imply $x, y \in S$.

- (II) ($\Rightarrow$): We'll prove that any elements in $A \setminus S$ was contained in a prime ideal which is disjoint to S : $\forall a \in A \setminus S, (a) \cap S \subseteq \{0\}$, otherwise if $a^n = a \cdot a^{n-1} \in S$, we'll have $a \in S$, contradictory.

Now we consider Σ be the set of all ideals containing (a) and being disjoint to S . By *Zorn's Lemma*, we can pick out a maximal element, namely $\mathfrak{p} \subseteq \Sigma$. We'll prove $\mathfrak{p}$ is prime:

Suppose we have $xy \in \mathfrak{p}, x, y \notin \mathfrak{p}$. Since $\mathfrak{p}$ is maximal and $\mathfrak{p} + (x), \mathfrak{p} + (y) \supsetneq \mathfrak{p}$, we know both of $\mathfrak{p} + (x), \mathfrak{p} + (y)$ intersects with S . Therefore $\exists s_1, s_2 \in S$ such that $\exists p_1, p_2 \in \mathfrak{p}, a_1, a_2 \in A$,

$$s_1 = p_1 + a_1 x, s_2 = p_2 + a_2 x$$

Then we have

$$(s_1 - p_1)s_2 \in S$$

whereas

$$\begin{aligned} (s_1 - p_1)s_2 &= a_1 x(p_2 + a_2 y) \\ &= a_1 p_2 x + a_1 a_2 xy \in \mathfrak{p} \end{aligned}$$

Which means $(s_1 - p_1)s_2 \in S \cap \mathfrak{p}$, contradicting with the construction of $\mathfrak{p}$.

□

- (ii) By (i), $\overline{S}$ defined by the complement in A of union of prime ideals disjoint from S is indeed a saturated multiplicative closed subset of A .

We only have to prove $\overline{S}$ is the unique minimal one, which is trivial since the family of sets, which consists of prime ideals disjoint from S , has a unique maximal element (final object, where morphisms taken under inclusion).

- (iii)

$$\mathfrak{p} \cap S \subsetneq \{0\}$$

$\Leftrightarrow$

$$\exists p \in \mathfrak{p}, a \in \mathfrak{a}, \text{ s.t. } 1 + p = a$$

$\Leftrightarrow$

$$\mathfrak{a} + \mathfrak{p} = (1)$$

$\Leftrightarrow$

$\mathfrak{a}, \mathfrak{p}$ are coprime

Conversely, it's easy to prove that $\mathfrak{a}, \mathfrak{p}$ being coprime will implies $\mathfrak{p} \cap S \subseteq \{0\}$. Therefore $\overline{S}$ is the complement of the union of all prime ideals which are not coprime with $\mathfrak{a}$.

Exercise 8

(UNFINISHED)

(I) $(i) \Rightarrow (ii)$: $\forall t \in T, \phi(\frac{t}{1}) = \frac{t}{1}$ is a unit in $T^{-1}A$. Since ϕ is bijective, the preimage of the inverse of $\frac{t}{1} \in T^{-1}A$ is also the inverse of $\frac{t}{1} \in S^{-1}A$.

(II) $(ii) \Rightarrow (iii)$: $\forall t \in T, \frac{t}{1}$ is a unit in $S^{-1}A$ and we denote $\frac{a}{s} = (\frac{t}{1})^{-1}$. Therefore $\exists s_0 \in S$, s.t. $s_0(at-s) = 0$, which indicate

$$(s_0a)t = s_0s \in S$$

Thus we can let $x = s_0a$.

(III) $(iii) \Rightarrow (iv)$: Suppose $\exists t \in T$, s.t. t is contained in a prime ideal $\mathfrak{p}$ which is disjoint from S . By (iii) , $\exists x \in A$, s.t. $xt \in S$. However, $t \in \mathfrak{p} \Rightarrow xt \in \mathfrak{p} \Rightarrow xt \in S \cap \mathfrak{p}$, contradictory.

(IV) $(iv) \Rightarrow (v)$: Otherwise T cannot be contained in $\overline{S}$.

(V) $(v) \Rightarrow (i)$:

Exercise 9

(I) We follow from the hint: Denote the family of all multiplicatively closed set which doesn't contains 0 by Σ , the problem is equivalent to prove that all the maximal elements in Σ must contains S .

We prove by contradictory and suppose there exist a maximal element $S \in \Sigma$ such that $\exists a \in S_0 \setminus S$. Let $S_a = \{sa^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$, which is a multiplicatively closed set strictly contains S . Because S is maximal in Σ , $S_a \notin \Sigma, 0 \in S_a$. Hence, $\exists n \in \mathbb{Z}^+$, s.t.

$$sa^n = 0$$

However, from $a \in S_0 \setminus S$ we know a^n is a non-zero-divisor, i.e. $s = 0$, which is absurd. Hence the proof was done. $\square$

(II) (i) Homomorphism $A \longrightarrow S_0^{-1}A$ is injective iff $\forall s \in S_0, a \in A \setminus \{0\}, sa \neq 0$. This is to say S_0 consists of non-zero-divisors.

(ii) Every element in $A \setminus S_0$ will be mapped to a zero-divisor while every element in S_0 will be mapped to a unit, obviously.

(iii) Injectivity is obtained from (i), and the preimage of $\frac{a}{s} \in S^{-1}A$ is $as^{-1} \in A$, obviously. $\square$

Remark 1. From **Exercise 3.6, 3.9**, we can see saturated sets and sets made up of prime-ideals-union acts as complements to each other. And if we let S be maximal, the corresponding $A \setminus S$ will be left to be only one prime idea.

Exercise 10